MIP Contest 2025/2026

An Automatic UNet-based Pipeline for Intima-Media Complex Segmentation in Carotid Ultrasound

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Introduction

The segmentation of the intima-media complex (IMC) in B-mode carotid ultrasound (US) images is a critical step for measuring the intima-media thickness (IMT), a key biomarker for cardiovascular risk assessment.

The objective of this work was to develop and evaluate a fully automatic pipeline for the segmentation of the Intima-Media Complex (IMC), from which IMT can be derived. The core method is based on the UNet architecture, a standard for biomedical image segmentation. Its performance was, then, evaluated on an internal validation set using the Dice Similarity Coefficient (DSC, Eq.(1)) and IMT measurement error ($Error_{IMT}$, Eq.(2)).

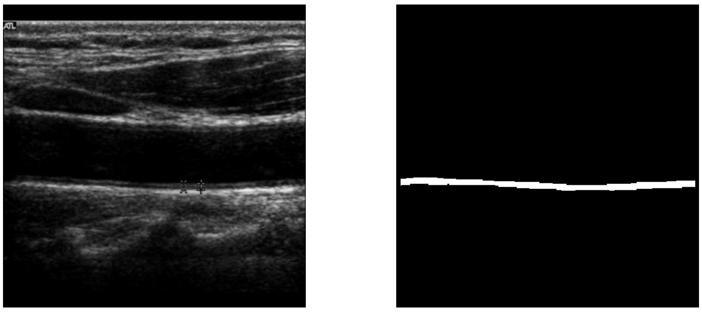
$$DSC(X, Y) = \frac{2|X \cap Y|}{|X| + |Y|} \tag{1}$$

$$Error_{IMT} = |IMT_{manual} - IMT_{automatic}| \tag{2}$$

where X represents the IMC target mask and Y is the mask predicted by our model.

Pipeline

The dataset provided by the contest consists of 2576 B-mode carotid ultrasound (US) images, each associated with a manual segmentation performed by an expert operator.



(a) (b)

Figure 1: Example of US image (a) and its corresponding binary IMC mask (b).

To develop and validate our model, the dataset was split into three independent subsets:

- **Training Set:** 1803 images (70 %)
- **Validation Set:** 386 images (15 %)
- **Test Set:** 387 images (15 %)

The training set was used to update model weights. The validation set was used during training to monitor for overfitting, save the best model weights, and optimize post-processing parameters. The test set was held out and used only for the final performance evaluation.

A high-level diagram of the developed pipeline is shown in Figure 2:

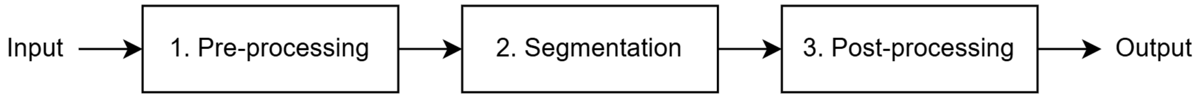
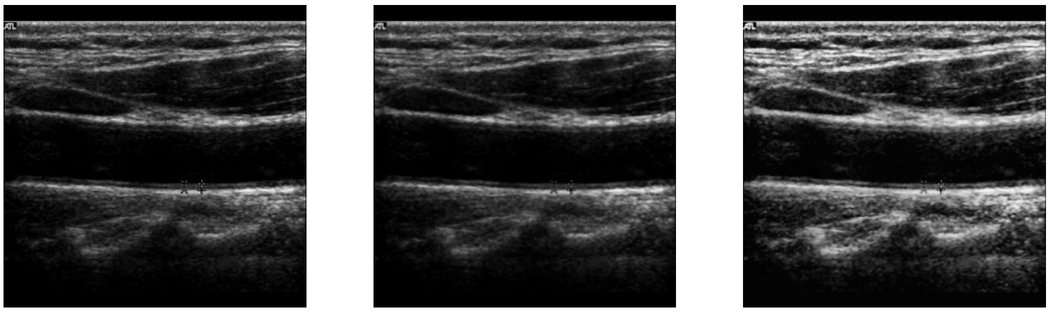


Figure 2

1. Pre-processing

Ultrasound (US) images inherently suffer from characteristic speckle noise. Therefore, the images required a pre-processing step for noise reduction, and several approaches were tested. It was found that a combination of a median filter and CLAHE (Contrast Limited Adaptive Histogram Equalization) over-amplified background noise and *decreased* performance. In contrast, using a **median filter** with kernel size 3 alone was the most effective strategy, as it successfully reduced speckle noise while preserving the boundaries of the thin IMC structure.



(a) Original Image (b) Median Filter (k=3) (c) Median Filter + CLAHE

Figure 3: Visual comparison of pre-processing methods. (a) The original US image with characteristic speckle noise. (b) The median filter effectively reduces speckle. (c) The combination with CLAHE demonstrates the over-amplification of background noise, which degraded performance.

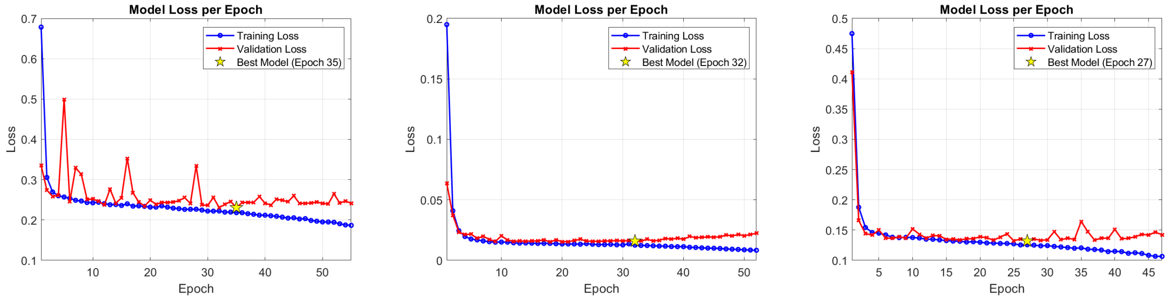
2. Segmentation

The core of the pipeline is a UNet, whose model was built using the PyTorch library and features a symmetric encoder-decoder path.

Input images were fed to the network at their original 480×480 pixel resolution. Common pre-processing steps like resizing or padding were deliberately avoided, as these operations could introduce artifacts or distort the thin IMC structure.

The encoder path progressively downsamples this 480×480 input via max pooling, while the decoder path uses bilinear upsampling to restore resolution. Crucially, skip connections concatenate feature maps from the encoder with the corresponding decoder levels, preserving high-resolution spatial information for precise localization. A final 1x1 convolution maps the features to a single-channel logit map for binary segmentation.

A systematic optimization was performed to define the final model parameters with a two-stage process. First, as loss function three options were tested: 'DiceLoss' (a), 'BCEWithLogitsLoss'(b), and a 50-50 hybrid of the two that was called 'CombinedLoss' (c), (Figure 4). While 'BCEWithLogitsLoss' achieved a fractionally higher peak validation DSC, its metrics were highly volatile and showed signs of overfitting. The 'CombinedLoss' provided the most stable validation loss and Dice score, demonstrating reliable convergence without volatility. Therefore, the 'CombinedLoss' was selected as it produced a more robust and trustworthy model.



(a) Dice Loss

(b) Binary Cross-Entropy Loss

(c) Combined Loss (BCE + Dice)

Figure 4: Evaluation of loss functions. While (b) BCEWithLogitsLoss achieved a high peak Dice Score, it showed significant volatility and overfitting. The (c) CombinedLoss (BCE + Dice) provided the most stable convergence and reliable validation metrics, and was thus selected for the final model.

Second, we optimized the remaining hyperparameters against a baseline model (depth=4, init_filters=32, learning_rate=1×10^{-3}) which used the 'CombinedLoss'. Variations of a single parameter at a time were tested: 'depth=5', 'init_filters=64', and 'learning_rate=1×10^{-4}'.

The results, summarized in Table 1, show that decreasing the learning rate did not improve performance. While increasing the filter count to 64 offered a marginal benefit, increasing the model depth to 5 showed the clearest improvement, achieving the highest validation Dice score of 0.7750.

Table 1: Hyperparameter optimization results on the validation set.

Parameter	Value Tested	Peak Validation DSC
Baseline	*(depth=4, init_filters=32, lr=1×10^{-3})*	**0.7691**
depth	5	**0.7750**
init_filters	64	0.7723
learning_rate	1×10^{-4}	0.7689

Therefore, the baseline architecture with 'CombinedLoss' and 'depth=5' was adopted for our final model. Training was performed using the Adam optimizer with a learning rate of 1×10^{-3} and a batch size of 8. An early stopping policy was implemented with a patience of 20 epochs, saving the model weights that achieved the highest mean Dice score on the validation set.

3. Post-processing

The raw output of the **UNet** is a probability map, which was converted into a final binary mask using a two-step post-processing approach.

First, an optimal threshold was determined by evaluating values from 0.1 to 0.9 on the validation set. This thresholding step alone achieved a mean validation Dice Score of 0.7703.

Second, various morphological sequences (using a 3x3 kernel) were tested on these masks to evaluate if "clean-up" operations could improve the score. The results showed that any sequence involving an 'opening' operation significantly degraded performance (e.g., 'only_opening' DSC = 0.7538), as this operation likely eroded the thin target IMC structure. In contrast, the 'only_closing' sequence achieved the highest score (Dice = 0.7704).

While the numerical improvement was marginal, this step was adopted as it was the best-performing operator and serves to fill small, unwanted holes in the final segmentation mask.

Results

The fully trained and optimized pipeline was evaluated on the test set. Performance was measured using the two metrics: Dice Similarity Coefficient (DSC) and the absolute IMT Error ($Error_{IMT}$).

The following table summarizes the performance of our method on the test set, showing the mean and standard deviation for both metrics. Our pipeline achieved a mean DSC of 0.7723 and a mean $Error_{IMT}$ of 1.342.

Table 2: Quantitative performance of the final pipeline on the internal test set.

Metric	**Mean**	**Std. Deviation**
Dice Similarity Coefficient (DSC)	0.7723	0.0842
$Error_{IMT}$	1.342	1.1825

Conclusion

The proposed pipeline, integrating a median filter, an optimized UNet, and post-processing, proved to be a robust strategy for IMC segmentation. The final model achieved strong quantitative performance on the test set, with a mean DSC of 0.7723 $\pm$ 0.0842 and a mean $Error_{IMT}$ of 1.342 $\pm$ 1.1825.

The systematic optimization process revealed several key findings: a simple median filter was superior to CLAHE for pre-processing; a CombinedLoss with a network depth of 5 provided the most stable and accurate segmentation; and a morphological closing step offered a slight improvement to the final masks. This work provides a validated baseline for automatic IMT measurement from B-mode ultrasound images.